

# **Lecture 17:**

# **Machine Learning Evaluation**

**CMSC/DATA 320: Introduction to Data Science**

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# Announcements

- PA4, WA4 due yesterday
  - PA5, WA5 out tomorrow
  - Quiz grades and solutions are posted a week after the quiz
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# **Model Evaluation**

# How do we know if our model is good?

In our KNN classification example, we ended by making a prediction.

How do we know if that prediction is any good?



# **Evaluation & Evaluation Metrics**



# Accuracy

The simplest way to check how well our model is doing is to look at its **accuracy**.

Accuracy is the **percentage of correct predictions**:

$$accuracy = \frac{\# \text{ correct predictions}}{\# \text{ total predictions}}$$

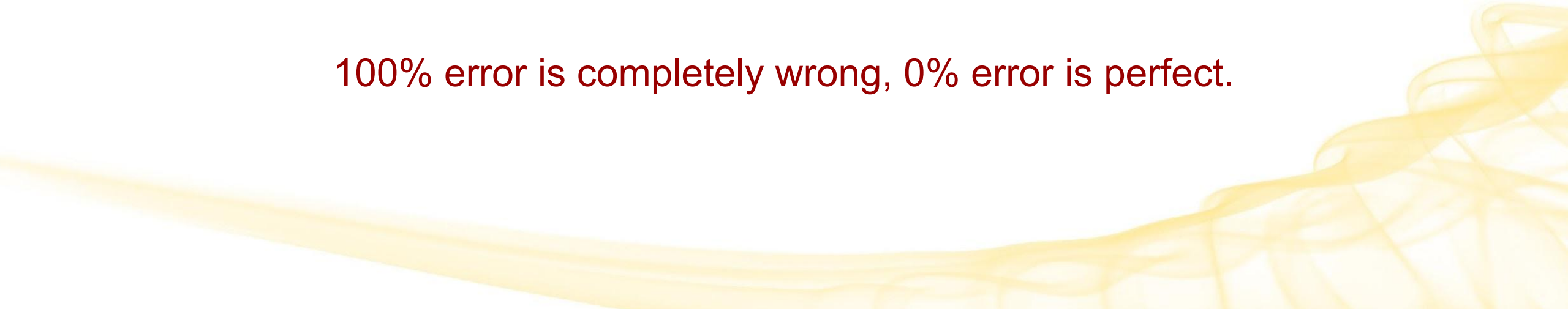
100% accuracy is perfect, 0% accuracy is completely wrong.

# Accuracy vs Error Rate

Error Rate is related to accuracy: it's the **percentage of incorrect predictions**:

$$\text{error rate} = 1 - \text{accuracy} = \frac{\# \text{ incorrect predictions}}{\# \text{ total predictions}}$$

100% error is completely wrong, 0% error is perfect.



# Accuracy Example

If 90 out of 100 predictions are correct for spam detection in an email classification model, what is the model's accuracy?

$$accuracy = \frac{\# \text{ correct predictions}}{\# \text{ total predictions}}$$

**Unfortunately, accuracy cannot always be used as the sole metric to evaluate model performance.**

# Accuracy Example

If 90 out of 100 predictions are correct for spam detection in an email classification model, what is the model's accuracy?

$$\begin{aligned} \text{accuracy} &= \frac{\# \text{ correct predictions}}{\# \text{ total predictions}} \\ &= 90/100 = 0.9 = 90\% \end{aligned}$$

**Unfortunately, accuracy cannot always be used as the sole metric to evaluate model performance.**

# Class Imbalance

Consider this scenario:

- Imagine a dataset with a categorical label that is either “positive” or “negative.”
- Suppose 90% of the categorical labels are negative, and 10% are positive.

Is 80% accuracy impressive here? What about 90%?



# Class Imbalance

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- Suppose 90% of the categorical labels are negative, and 10% are positive.

Is 80% accuracy impressive here? What about 90%?

This skewed divide is called a **Class Imbalance**.

# Class Imbalance

Majority class

Minority class

Consider this scenario:

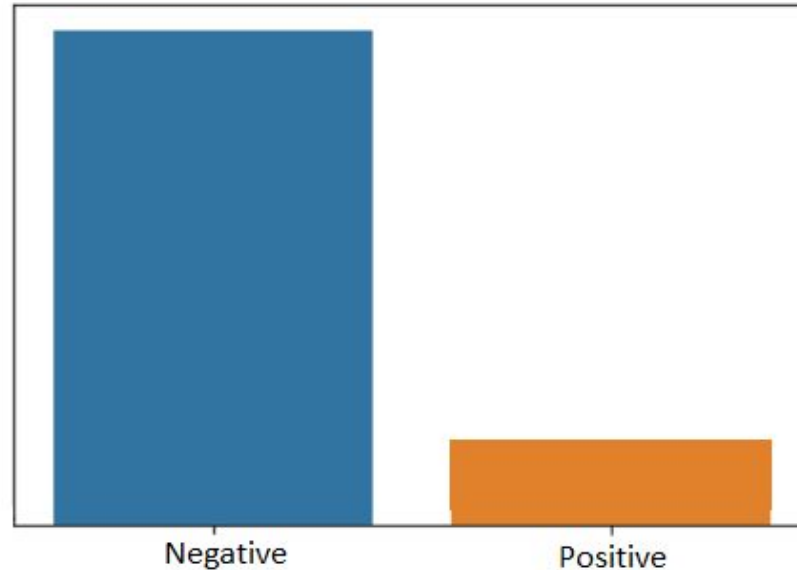
- Imagine a dataset with a categorical label that is either “positive” or “negative.”
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Is 80% accuracy impressive here? What about 90%?

This skewed divide is called a **Class Imbalance**.

# Class Imbalance: A serious problem

Class imbalance happens when an **overwhelming majority** of your **data is a single class/category** and requires care.



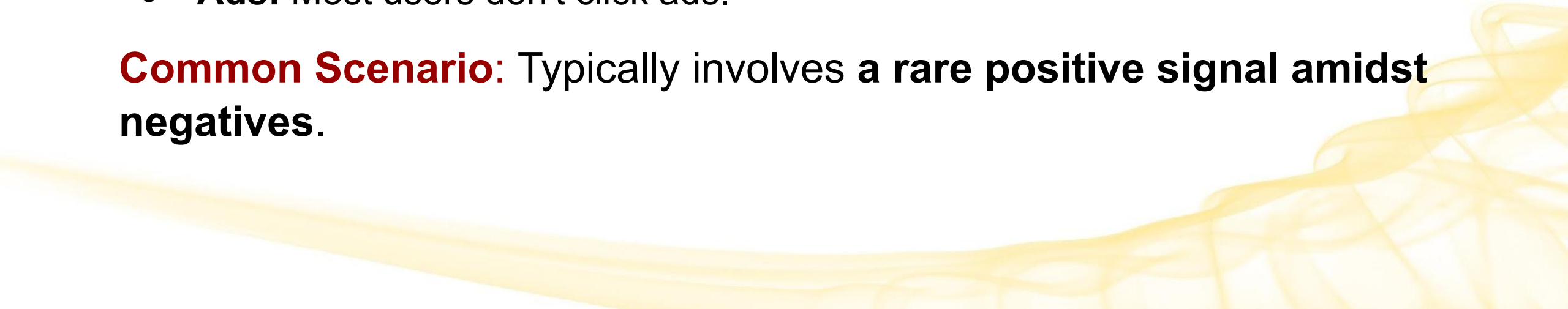
# Situations With Class Imbalance

**Class Imbalance Issue:** Occurs when **one class dominates the data**, making one group significantly larger than others and causing skew/bias.

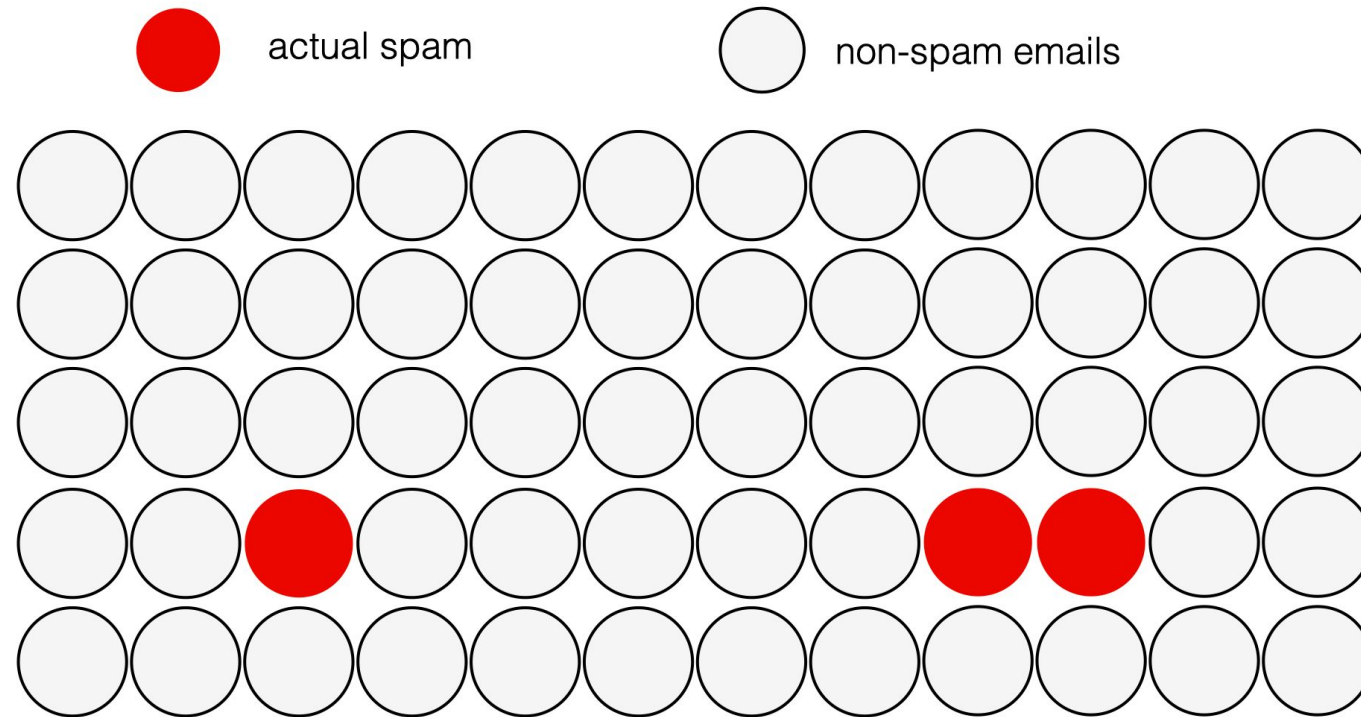
## Examples:

- **Fraud:** Most transactions are not fraudulent.
- **Disease:** The majority of people don't have cancer.
- **Ads:** Most users don't click ads.

**Common Scenario:** Typically involves a **rare positive signal amidst negatives**.



# Class Imbalance Visualized



# Limitation of the Accuracy metric

Accuracy may not show the full model performance, especially when classes are imbalanced.

**Issue with Imbalanced Data:** Accuracy can mislead when a dataset is imbalanced and one class dominates.

- *High accuracy might be achieved* by simply predicting the majority class, even if the model is performing poorly on the minority class.

***Accuracy fails to be a reliable measure when class distribution is skewed.***

# Limitation of the Accuracy metric

## Neglecting Error Types:

Accuracy **doesn't differentiate** between false positives and false negatives, treating all errors or misclassification equally.

- False negatives and false positives may not be equally bad!

**Alternative evaluation metrics are needed.**

# Confusion Matrix

A matrix that **summarizes the performance** of a machine learning model on a set of test data.

- Shows true positives, true negatives, false positives, and false negatives
  - TP, TN, FP, FN
- Offers insights into the model's accuracy and prediction errors across classes.

***Tells us how wrong our model is about different things.***

# Building a Confusion Matrix

Let's go back to our “positive” or “negative” categorical example.  
(2-class classification)

Instead of looking at our model accuracy, let's look at our true positive, false positive, true negative, and false negative (TP, FP, FN, TN) rate:

|                     | Positive | Negative |
|---------------------|----------|----------|
| Classified Positive | 80%      | 0%       |
| Classified Negative | 10%      | 10%      |

# Building a Confusion Matrix

Let's go back to our "positive" or "negative" categorical example. (2-class classification)

Instead of looking at our model accuracy, let's look at our true positive, false positive, true negative, and false negative (TP, FP, FN, TN) rate:

|                     | Positive | Negative |
|---------------------|----------|----------|
| Classified Positive | 80%      | 0%       |
| Classified Negative | 10%      | 10%      |



|                    | Actual Positive               | Actual Negative               |
|--------------------|-------------------------------|-------------------------------|
| Predicted Positive | ✓ True Positive<br><b>TP</b>  | ⚠ False Positive<br><b>FP</b> |
| Predicted Negative | ⚠ False Negative<br><b>FN</b> | ✓ True Negative<br><b>TN</b>  |

# Confusion Matrix Example

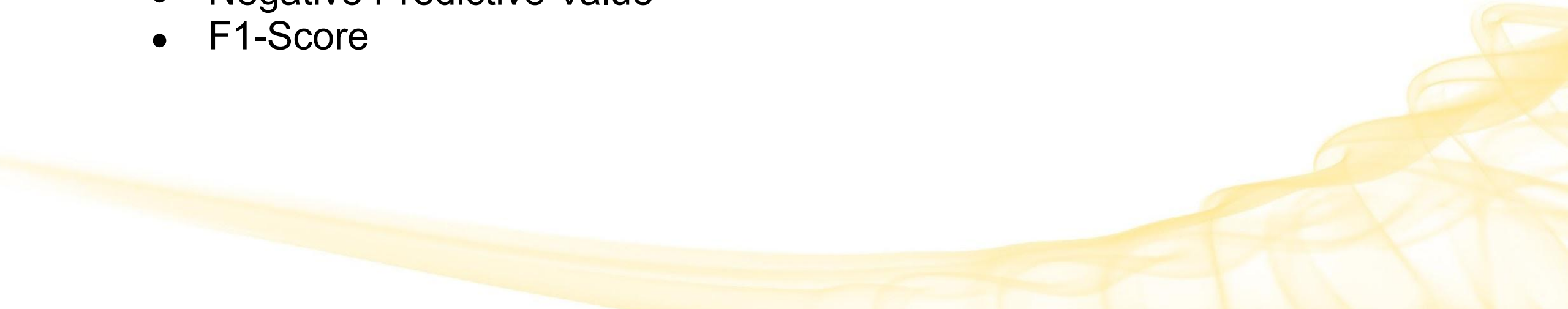
SPAM classification

|                 | Actual isSpam = True                                   | Actual isSpam = False                                      |
|-----------------|--------------------------------------------------------|------------------------------------------------------------|
| Predicted True  | ✓ Email is SPAM and we predicted is SPAM<br><b>TP</b>  | ⚠ Email is not SPAM but we predicted is SPAM<br><b>FP</b>  |
| Predicted False | ⚠ Email is SPAM but we predicted not SPAM<br><b>FN</b> | ✓ Email is not SPAM and we predicted not SPAM<br><b>TN</b> |

# Confusion Matrix Flexibility

A confusion matrix can be used to calculate various evaluation metrics.

## Examples:

- Accuracy
  - Precision
  - Recall (Sensitivity or True Positive Rate)
  - Specificity (True Negative Rate)
  - Negative Predictive Value
  - F1-Score
- 

# Confusion Matrix Metrics

|                    | Actual Positive                                     | Actual Negative                            |                                                          |
|--------------------|-----------------------------------------------------|--------------------------------------------|----------------------------------------------------------|
| Predicted Positive | ✓ Correct Answer!<br><b>TP</b>                      | ⚠ Type I Error<br><b>FP</b>                | <b>Precision</b><br>$\frac{TP}{TP + FP}$                 |
| Predicted Negative | ⚠ Type II Error<br><b>FN</b>                        | ✓ Correct Answer!<br><b>TN</b>             | <b>Negative Predictive Value</b><br>$\frac{TN}{TN + FN}$ |
|                    | <b>Recall (Sensitivity)</b><br>$\frac{TP}{TP + FN}$ | <b>Specificity</b><br>$\frac{TN}{TN + FP}$ | <b>Accuracy</b><br>$\frac{TP + TN}{TP + TN + FP + FN}$   |

# Precision

How **precise** we are in identifying positive cases



# Precision

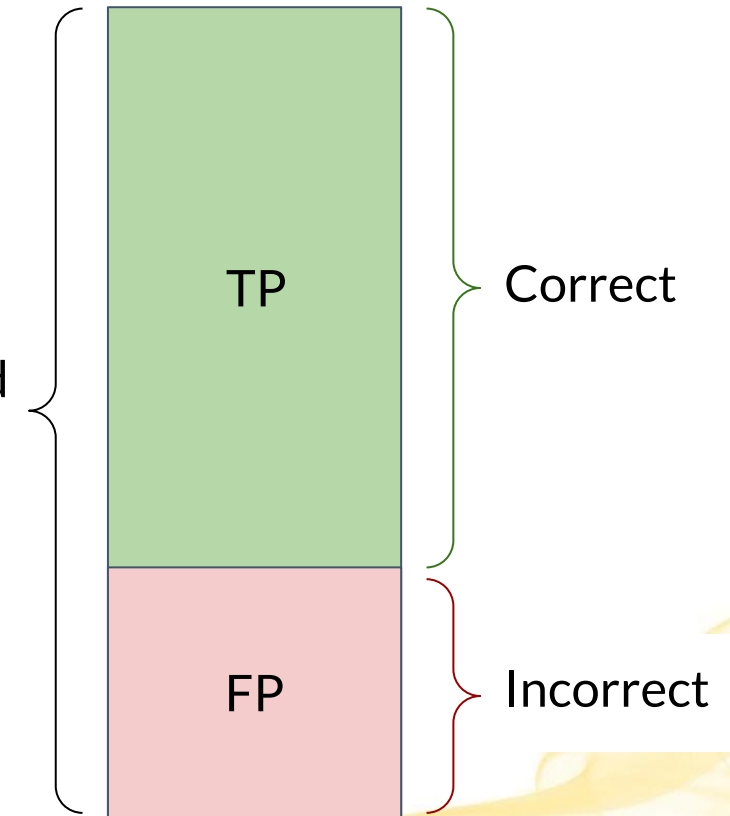
How **precise** we are in identifying **positive cases**.

Precision measures how many of the things that we classified as positive were actually positive.

|                    | Actual Positive                | Actual Negative             |                                          |
|--------------------|--------------------------------|-----------------------------|------------------------------------------|
| Predicted Positive | ✓ Correct Answer!<br><b>TP</b> | ⚠ Type I Error<br><b>FP</b> | <b>Precision</b><br>$\frac{TP}{TP + FP}$ |

$$precision = \frac{TP}{TP + FP} = \frac{\text{actually positive}}{\text{all predicted positive}}$$

Everything we predicted would be positive.



# Measuring Precision

- Measured on a scale of 0 to 1, or as a percentage.
  - **Higher precision is better.**
  - Value 1.0 indicates the model is always correct when predicting the target class (positive case) and *never makes a mistake if it predicts positive.*
- 

# Precision Example

|                   | Predicted<br>1 – Spam | Predicted<br>0 – Ham |                         |
|-------------------|-----------------------|----------------------|-------------------------|
| Actual<br>1 -Spam | 90 [TP]               | 1 [FN]               | → Spam Emails [90+1=91] |
| Actual<br>0 – Ham | 4 [FP]                | 5 [TN]               | → Good Emails [ 4+5 =9] |

$$\text{Precision} = \frac{90}{90+4} = 0.95$$

Precision = 95%

## Precision

- Measured on a scale of 0 to 1, or as a percentage.
- **Higher precision is better.**
- Value 1.0 indicates the model is always correct when predicting the target class and *never makes a mistake if it predicts positive.*

Note: matrix columns and rows transposed here – not uncommon.

# When to Use Precision

We use precision when **we only care about being correct about the things we identify as positive.**

- Ex. Many big tech firms do not care if they turn away 1000 great engineers; they only care about making sure the ones they DO hire are great.
  - False negatives are acceptable, but false positives are not
- Ex. Suppose we have a mushroom dataset
  - 1 = a mushroom is edible, 0 = a mushroom is poisonous
  - FP: the mushroom is poisonous, but the model predicted it is edible
  - FN: the mushroom is edible, but the model predicted it is poisonous
  - Which is worse: the FP or the FN?

**Precision answers the question: *how often are the positive predictions correct?***

# Recall

How good we are at **detecting/identifying** all actual positive cases



# Recall (Sensitivity, or True Positive Rate [TPR])

A measure of **how many positives** your model is able to **recall (find)** in the data.

|                    |                                              |
|--------------------|----------------------------------------------|
|                    | Actual Positive                              |
| Predicted Positive | ✓ Correct Answer!<br>TP                      |
| Predicted Negative | ⚠ Type II Error<br>FN                        |
|                    | Recall (Sensitivity)<br>$\frac{TP}{TP + FN}$ |

**Recall:** The proportion of true positives to the total actual positives.  
Recall measures the model's ability to *identify positive instances*.

$$recall = \frac{TP}{TP + FN}$$

- Measured on a scale of 0 to 1 or as a percentage.
- **Higher recall is preferable.**
- Value of 1.0 means the model captures all instances of the target class (positive case) and never misses it in predictions.

# Recall Example

|                         | Predicted<br>1 –Cancer | Predicted<br>0 – No Cancer |
|-------------------------|------------------------|----------------------------|
| Actual<br>1 -Cancer     | 90 [TP]                | 4 [FN]                     |
| Actual<br>0 – No Cancer | 1 [FP]                 | 5 [TN]                     |

→ Cancer Records [90+4=94]

→ Non Cancer Records [ 5+1 =6]

$$Recall = \frac{90}{90 + 4} = 0.95$$

Recall = 95%

$$recall = \frac{TP}{TP + FN}$$

## Recall

- Measured on a scale of 0 to 1, or as a percentage.
- **Higher recall is better.**
- Value of 1.0 means the model captures all instances of the target class and never misses it in predictions.

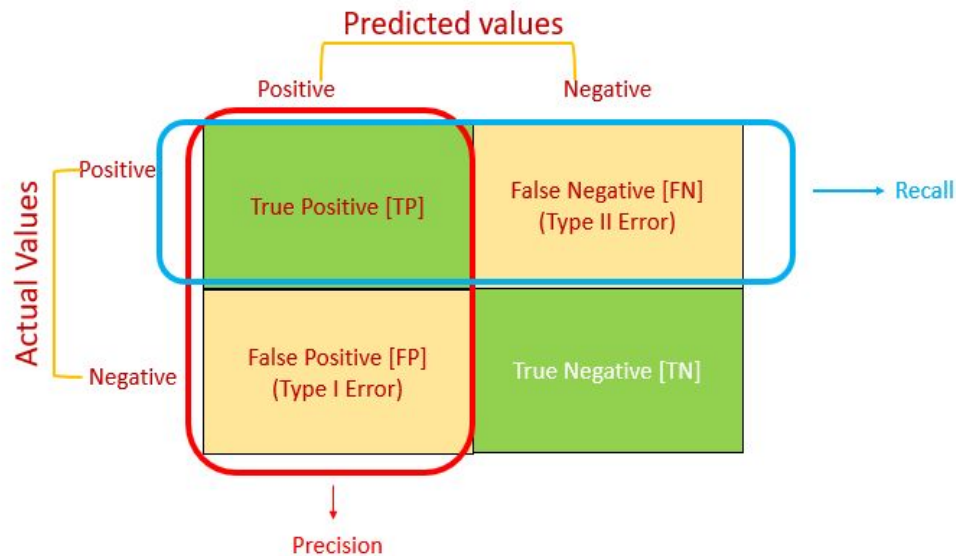
# When to use Recall:

When we want to make sure ***we don't miss any positive cases.***

- Example: identifying people contagious with a deadly super-plague
  - 1 = they have the plague
  - 0 = they do not have the plague
  - FP: they do not have the plague, but we predicted they do
  - FN: they do have the plague, but we predicted they do not
  - Which is worse, FP or FN?
    - FN are not acceptable here because we would miss them having a contagious, deadly super-plague!
    - Better to have a FP and we do further testing

# Precision vs. Recall

In the given scenarios, which metric (precision or recall) holds more practical importance or application?



- Testing for cancer
- Fraud alerts
- Loans

Think about:

**Precision:** Aims to be right when it says something is positive (minimize false positives).

**Recall:** Aims to not miss anything that's actually positive (minimize false negatives).

# Precision vs. Recall Tradeoff

In binary classification (what we've been discussing), you can't simultaneously have both high precision and high recall.

- If you increase precision, you decrease recall
- If you increase recall, you decrease precision

Example: Baseball players 

- **High precision player:** Takes fewer swings because wants to make sure to hit them
  - Averse to FPs
  - Doesn't take a lot of swings, but when they do they hit (TP)
- **High recall player:** Takes more swings because wants to take advantage of any chance to get a hit
  - Averse to FNs
  - Takes lots of swings, so misses more (FPs) but hits more (TP), too

# Confusion Matrix Metrics

How precise we are in identifying positive cases

|                    | Actual Positive                                     | Actual Negative                            |                                                        |
|--------------------|-----------------------------------------------------|--------------------------------------------|--------------------------------------------------------|
| Predicted Positive | ✓ Correct Answer!<br><b>TP</b>                      | ⚠ Type I Error<br><b>FP</b>                | <b>Precision</b><br>$\frac{TP}{TP + FP}$               |
| Predicted Negative | ⚠ Type II Error<br><b>FN</b>                        | ✓ Correct Answer!<br><b>TN</b>             | <b>NPV</b><br>$\frac{TN}{TN + FN}$                     |
|                    | <b>Recall (Sensitivity)</b><br>$\frac{TP}{TP + FN}$ | <b>Specificity</b><br>$\frac{TN}{TN + FP}$ | <b>Accuracy</b><br>$\frac{TP + TN}{TP + TN + FP + FN}$ |

How many positives the model is able to recall from the data

# Dealing with Precision vs. Recall Tradeoff

Sometimes, we wish for a model to have as few FPs and FNs as possible  
→ we want to **maximize both precision and recall**.

- Example: Identifying Harmful Content
  - Goal is to ensure that flagged content is both correct (high precision) and that no inappropriate content is overlooked (high recall).
- In practice, it is not possible to completely maximize both because of the trade-off between precision and recall.
- **What can we do?**
  - Is there some way to combine precision and recall into a single thing that we can optimize, that balances the two?

# F1 Score

$$\text{Harmonic mean } H(x_1, x_2, \dots, x_n) = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}.$$

F1 score is a metric is used when you seek a **balance between precision and recall** for a machine learning model.

- Harmonic mean of precision and recall
  - Used when both are important
  - Harmonic mean averages reciprocal rates
- Range between 0 and 1
  - The closer to 1, the better the model (in this regard of balancing precision and recall)

$$\begin{aligned} \text{F1 Score} &= \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} \\ &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\ &= \frac{TP}{TP + \frac{1}{2}(FP + FN)} \end{aligned}$$

# F1 Score

## Pros

- Accounts for class imbalance
- Gives us a way to optimize on a combination of precision and recall

## Cons

- Harder to interpret
- Assumes precision and recall are equally important
  - Not always the case!
- Doesn't take into account error distribution or prediction confidence

Recommendation: use in combination with other metrics

# Metrics Summary

**Precision:** Focuses on minimizing false positives in predictions.

**Recall:** Focuses on minimizing false negatives in predictions.

**F1 Score:** Balances both precision and recall in a single metric. Useful for scenarios where false positives and false negatives are equally important.

